

Laird Stewart

hlairdstew@gmail.com lairdstewart.com



EDUCATION

M.S. Computer Science, New York University 8/26–5/28

Johns Hopkins Engineering for Professionals (non-degree) 1/24–12/25

Theory of Machine Learning, Monte Carlo Methods, Electromagnetics, Fluid Dynamics, Plasma Physics

B.S. Computer Engineering, Georgia Institute of Technology 8/21–5/23

4.0 GPA, Tau Beta Pi (Engineering honor society), Eta Kappa Nu (IEEE honor society)

B.S. Applied Math and Stats, Emory University 8/18–5/21

3.9 GPA, Minor in Economics, Omicron Delta Epsilon (Economics Honor Society)

Through Emory's Dual Degree Engineering Program, earned separate bachelor's degrees from Emory and Georgia Tech. Completed Emory degree requirements in three years (8/18–5/21) then transferred to Tech to complete engineering degree using transfer credits from Emory. Both degrees conferred 5/23.

WORK EXPERIENCE

Software Engineer, Metron 6/23–7/26

Metron builds software grounded in applied math research for the Defense and Intelligence Community

- Core developer on TRK, the U.S. Navy's premier data fusion and tracking tool for anti-submarine warfare
- Implemented forward-backward particle smoothing feature in TRK, from algorithm to data visualization
- Lead engineer on ONR project to jointly estimate target submarine and environment state using acoustic-ray tracing. Leveraged normalizing flows and "ensemble samplers with affine invariance" for performance
- Organized a particle filter literature review group and a book club for Bishop's *Deep Learning* textbook

Data Science Intern, Equifax 5/21–8/21

- Built and automated end-to-end training pipeline for random forest classifier of U.S. households in R/SQL

Data Science Consultant, ERASE Child Trafficking 2/21–3/21

- Placed first in an Emory data science competition to help grow ERASE's Facebook outreach
- Built Linear Regression and Random Forest models on 2500 posts in Python to predict successful posts. Presented a 10-minute summary to ERASE leadership on how to increase engagement

Wealth and Investment Management Intern, Wells Fargo 7/20–8/20

- Researched six public equities and built discounted cash flow valuations to support portfolio composition

Tutor, Emory University 1/19–2/20

- Tutored Emory undergraduates in linear algebra and logic (philosophy) in 50 minute, 1-on-1 sessions

PUBLICATION

SIR Model of Time-Dependent Drug and Vaccine Distribution on COVID-19 4/21

Published in SIAM Undergraduate Research Online

- Expanded a standard "Susceptible-Infected-Recovered" (SIR) system of differential equations to include the effects of drug/vaccine deployment and modeled them in Python. Led team of four undergraduates.

ADDITIONAL INFORMATION

Skills: Java, Python, Linux, OpenGL, Protobuf/gRPC, Guice, Concurrent/Parallel programming in Java

Hobbies: Tennis, Squash, Running, Rock Climbing, Skiing, Backpacking, Hiking: summited 12 14,000 ft peaks